

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Tutorial 2: Solutions

Date: Aug 19, 2023

1. Refer to the electric circuit shown in Figure 1. Perform *Nodal Analysis* and express the node voltages (w.r.t. reference node R) in the compact $V = [v_a \ v_b \ v_c \ v_d]^T = Y^{-1}CU$ form (usual notations). Use the following numerical values and evaluate the matrix Y and vector CU . $R_1 = 20\Omega$, $R_2 = 40\Omega$,

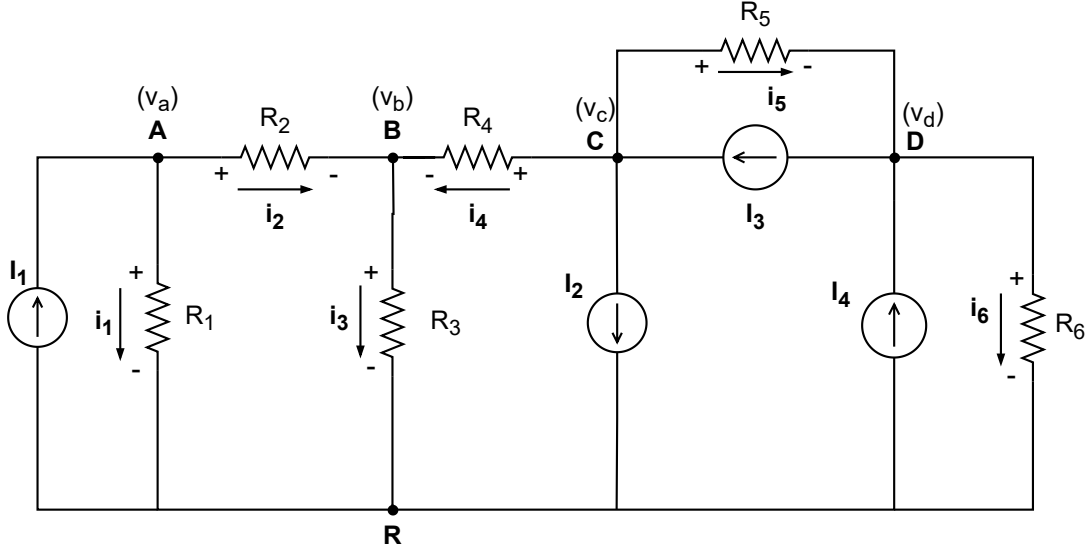


Figure 1 Electric circuit with independent current sources.

$R_3 = 100\Omega$, $R_4 = 50\Omega$, $R_5 = 10\Omega$, $R_6 = 200\Omega$, $I_1 = 10A$, $I_2 = 2.5A$, $I_3 = 2A$, $I_4 = 5A$.

Solution

Apply KCL at node A:

$$\begin{aligned} \frac{V_a}{R_1} + \frac{V_a - V_b}{R_2} &= I_1 \\ \Rightarrow (G_1 + G_2)V_a - (G_2)V_b &= I_1 \end{aligned} \quad (1)$$

Apply KCL at node B:

$$\begin{aligned} \frac{V_a - V_b}{R_2} + \frac{V_c - V_b}{R_4} - \frac{V_b}{R_3} &= 0 \\ \Rightarrow -(G_2)V_a + (G_2 + G_4 + G_3)V_b - (G_4)V_c &= 0 \end{aligned} \quad (2)$$

Apply KCL at node C:

$$\begin{aligned} \frac{V_c - V_b}{R_4} + I_2 + \frac{V_c - V_d}{R_5} &= I_3 \\ \Rightarrow -(G_4)V_b + (G_4 + G_5)V_c - (G_5)V_d &= -I_2 + I_3 \end{aligned} \quad (3)$$

Apply KCL at node D:

$$\begin{aligned} I_3 + \frac{V_d}{R_6} &= I_4 + \frac{V_c - V_d}{R_5} \\ \Rightarrow -(G_5)V_c + (G_5 + G_6)V_d &= -I_3 + I_4 \end{aligned} \quad (4)$$

From the equations (1), (2), (3) and (4), we can write the $YV = CU$ form as follows

$$Y = \begin{bmatrix} G_1 + G_2 & -G_2 & 0 & 0 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 & 0 \\ 0 & -G_4 & G_4 + G_5 & -G_5 \\ 0 & 0 & -G_5 & G_5 + G_6 \end{bmatrix} \quad V = \begin{bmatrix} V_a \\ V_b \\ V_c \\ V_d \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \quad U = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix}$$

The numerical values of the resistors are as follows.

$$R_1 = 20, R_2 = 40, R_3 = 100, R_4 = 50, R_5 = 10, R_6 = 200.$$

Thus, the numerical values of the conductance are as follows.

$$G_1 = 0.05, G_2 = 0.025, G_3 = 0.01, G_4 = 0.02, G_5 = 0.1, G_6 = 0.005$$

The numerical values of the current sources are as follows.

$$I_1 = 10A, I_2 = 2.5A, I_3 = 2A, I_4 = 5A$$

$$Y = \begin{bmatrix} 0.075 & -0.025 & 0 & 0 \\ -0.025 & 0.055 & -0.02 & 0 \\ 0 & -0.02 & 0.12 & -0.1 \\ 0 & 0 & -0.1 & 0.105 \end{bmatrix}$$

$$C * U = \begin{bmatrix} 10 \\ 0 \\ -0.5 \\ 3 \end{bmatrix}$$

2. Refer to the electric circuit shown in Figure 2. The node voltages V_a , V_b , V_c and V_d are w.r.t node R . Construct the equation $YV = CU$ in terms of the circuit element values.

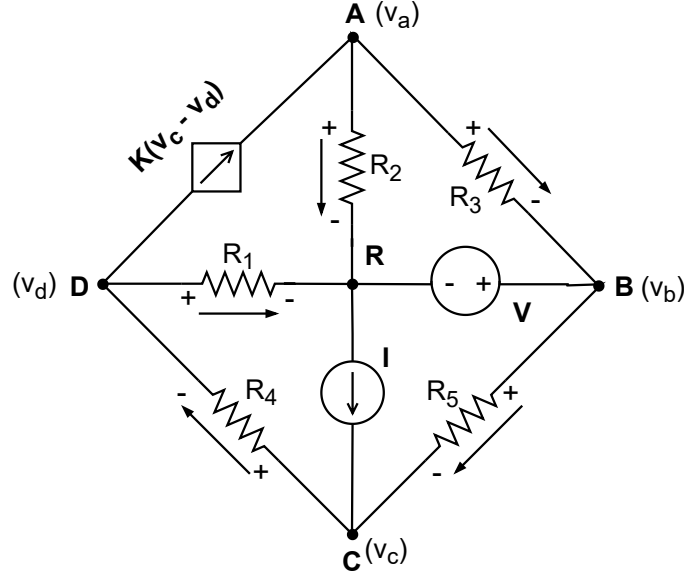


Figure 2 Electric circuit with both independent and dependent sources.

Solution Apply KCL at Node A

$$\begin{aligned}\frac{v_a}{R_2} + \frac{v_a - v_b}{R_3} &= k(v_c - v_d) \\ \rightarrow G_2 v_a + G_3 v_a - k v_c + k v_d &= G_3 V \quad [v_b = V] \\ \rightarrow (G_2 + G_3) v_a - k v_c + k v_d &= G_3 V\end{aligned}$$

Apply KCL at Node C

$$\begin{aligned}\frac{v_b - v_c}{R_5} + I &= \frac{v_c - v_d}{R_4} \\ \Rightarrow G_5 V - G_5 v_c - G_4 v_c + G_4 v_d &= -I \quad [v_b = V] \\ \Rightarrow (G_4 + G_5) v_c - G_4 v_d &= I + G_5 V\end{aligned}$$

Apply KCL at Node D

$$\begin{aligned}\frac{v_c - v_d}{R_4} &= k(v_c - v_d) + \frac{v_d}{R_1} \\ \Rightarrow -(G_4 - k) v_c + (G_1 + G_4 - k) v_d &= 0\end{aligned}$$

$$\begin{bmatrix} G_2 + G_3 & -k & k \\ 0 & G_4 + G_5 & -G_4 \\ 0 & -(G_4 - k) & (G_1 + G_4 - k) \end{bmatrix} \begin{bmatrix} v_a \\ v_c \\ v_d \end{bmatrix} = \begin{bmatrix} 0 & G_3 \\ 1 & G_5 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} I \\ V \end{bmatrix}$$

3. The majority function $M(X, Y, Z)$ is equal to 1 only when two or three of its arguments equal 1. Consequently, the function is equal to 0 when two or three of its arguments equal 0.
- Express the function $M(X, Y, Z)$ as a sum of minterms (Standard SOP).
 - Express the function $M(X, Y, Z)$ as a product of maxterms (Standard POS).
 - Show that these expressions can be simplified as

$$M(X, Y, Z) = X.Y + X.Z + Y.Z = (X + Y)(X + Z)(Y + Z)$$

- Argue that $M(A, B, M(C, D, E)) = M(M(A, B, C), D, M(A, B, E))$.
- For the function $F(A, B, C, D)$ corresponding to the network of Figure 3, find a Boolean expression.

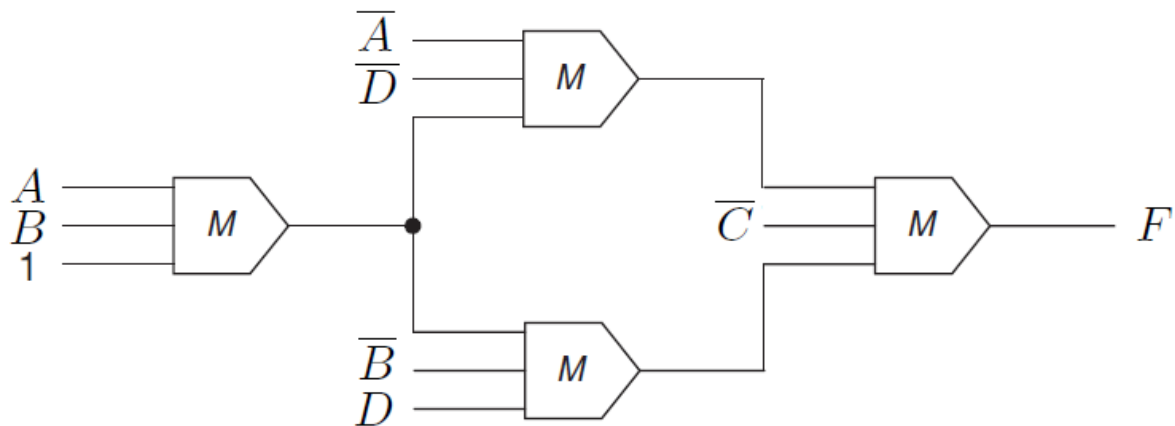


Figure 3 Digital circuit built using the majority function

- Simplify the expressions obtained in the previous step.

Solution

The truth table for the majority function is

X	Y	Z	M
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

- The function $M(X, Y, Z)$ can be written in the sum of minterms (Standard SOP) form as $M(X, Y, Z) = \overline{X}.Y.Z + X.\overline{Y}.Z + X.Y.\overline{Z} + X.Y.Z$.
- The function $M(X, Y, Z)$ can be written in the product of maxterms (Standard POS) form as $M(X, Y, Z) = (X + Y + Z) \cdot (X + Y + \overline{Z}) \cdot (X + \overline{Y} + Z) \cdot (\overline{X} + Y + Z)$
- The simplifications can be done using the identities $A + A = A$, $A.A = A$ and the distributive laws as follows:

$$\begin{aligned}
M(X, Y, Z) &= \overline{X}.Y.Z + X.\overline{Y}.Z + X.Y.\overline{Z} + X.Y.Z \\
&= \overline{X}.Y.Z + X.Y.Z + X.\overline{Y}.Z + X.Y.Z + X.Y.\overline{Z} + X.Y.Z \\
&= (\overline{X} + X).Y.Z + X.(\overline{Y} + Y).Z + X.Y.(\overline{Z} + Z) \\
&= X.Y + X.Z + Y.Z
\end{aligned}$$

$$\begin{aligned}
M(X, Y, Z) &= (X + Y + Z).(X + Y + \overline{Z}).(X + \overline{Y} + Z).(\overline{X} + Y + Z) \\
&= (X + Y + Z).(X + Y + \overline{Z}).(X + Y + Z).(\overline{X} + \overline{Y} + Z) \\
&\quad .(X + Y + Z).(\overline{X} + Y + Z) \\
&= (X + Y + Z.\overline{Z}).(X + Y.\overline{Y} + Z).(X.\overline{X} + Y + Z) \\
&= (X + Y)(X + Z)(Y + Z)
\end{aligned}$$

(d) We need to show that $M(A, B, M(C, D, E)) = M(M(A, B, C), D, M(A, B, E))$. We can consider the following cases.

Case I: $A = 0, B = 0$. Here $M(0, 0, M(C, D, E)) = 0$ and $M(M(0, 0, C), D, M(0, 0, E)) = M(0, D, 0) = 0$. This establishes the required result.

Case II: $A = 1, B = 1$. Here $M(1, 1, M(C, D, E)) = 1$ and $M(M(1, 1, C), D, M(1, 1, E)) = M(1, D, 1) = 1$. This establishes the required result.

Case III: $A = 1, B = 0$. If $M(C, D, E) = 0$, then at least two of C, D, E are 0. This implies at least two of $M(1, 0, C), D, M(1, 0, E)$ are 0. This establishes the required result. If $M(C, D, E) = 1$, then at least two of C, D, E are 1. This implies at least two of $M(1, 0, C), D, M(1, 0, E)$ are 1. This again establishes the required result.

Case IV: $A = 0, B = 1$. This is same as Case III as all the involved expressions are symmetric in A and B .

The same can be done using Boolean algebra as follows:

$$\begin{aligned}
M(A, B, M(C, D, E)) &= A.B + (A + B).M(C, D, E) \\
&= A.B + (A + B).(C.D + C.E + D.E) \\
&= A.B + (A + B).D.(C + E) + (A + B).C.E
\end{aligned}$$

$$\begin{aligned}
M(M(A, B, C), D, M(A, B, E)) &= D.(M(A, B, C) + M(A, B, E)) \\
&\quad + M(A, B, C).M(A, B, E) \\
&= D.((A.B + A.C + B.C) + (A.B + A.E + B.E)) \\
&\quad + (A.B + A.C + B.C).(A.B + A.E + B.E) \\
&= D.(A.B + (A + B).(C + E)) \\
&\quad + A.B.A.B + A.B.A.E + A.B.B.E \\
&\quad + A.C.A.B + A.C.A.E + A.C.B.E \\
&\quad + B.C.A.B + B.C.A.E + B.C.B.E \\
&= A.B.D + (A + B).D.(C + E) \\
&\quad + A.B + (A + B).C.E \\
&= A.B + (A + B).D.(C + E) + (A + B).C.E
\end{aligned}$$

- (e) A Boolean expression for the function $F(A, B, C, D)$, corresponding to the network of Figure 3 can be obtained as follows:

$$\begin{aligned} M(A, B, 1) &= (A + B)(1 + A)(1 + B) \\ &= (A + B) \end{aligned}$$

$$\begin{aligned} M(\overline{A}, \overline{D}, M(A, B, 1)) &= M(\overline{A}, \overline{D}, A + B) \\ &= (\overline{A} + (A + B))(\overline{D} + (A + B))(\overline{A} + \overline{D}) \\ &= (1 + B)(\overline{D} + (A + B))(\overline{D} + \overline{A}) \\ &= (\overline{D} + (A + B))(\overline{D} + \overline{A}) \\ &= \overline{D} + (A + B)\overline{A} \\ &= \overline{D} + A.\overline{A} + B.\overline{A} \\ &= \overline{A}.B + \overline{D} \end{aligned}$$

$$\begin{aligned} M(\overline{B}, D, M(A, B, 1)) &= M(\overline{B}, D, (A + B)) \\ &= (\overline{B} + (A + B))(D + (A + B))(\overline{B} + D) \\ &= (1 + A)(D + (A + B))(\overline{B} + D) \\ &= (D + (A + B))(D + \overline{B}) \\ &= D + (A + B)\overline{B} \\ &= D + A.\overline{B} + B.\overline{B} \\ &= A.\overline{B} + D \end{aligned}$$

$$\begin{aligned} F(A, B, C, D) &= M(\overline{A}.B + \overline{D}, \overline{C}, A.\overline{B} + D) \\ &= (\overline{C} + A.\overline{B} + D)(\overline{C} + \overline{A}.B + \overline{D})(A.\overline{B} + D + \overline{A}.B + \overline{D}) \end{aligned}$$

- (f) The Boolean expression for the function $F(A, B, C, D)$ can be simplified as follows:

$$\begin{aligned} F(A, B, C, D) &= M(\overline{A}.B + \overline{D}, \overline{C}, A.\overline{B} + D) \\ &= (\overline{C} + A.\overline{B} + D)(\overline{C} + \overline{A}.B + \overline{D})(A.\overline{B} + D + \overline{A}.B + \overline{D}) \\ &= (\overline{C} + A.\overline{B} + D)(\overline{C} + \overline{A}.B + \overline{D})(A.\overline{B} + \overline{A}.B + 1) \\ &= (\overline{C} + A.\overline{B} + D)(\overline{C} + \overline{A}.B + \overline{D}) \\ &= \overline{C} + (A.\overline{B} + D)(\overline{A}.B + \overline{D}) \\ &= \overline{C} + (A.\overline{B}.\overline{A}.B + D.\overline{D} + \overline{A}.B.D + A.\overline{B}.\overline{D}) \\ &= \overline{C} + (0 + 0 + \overline{A}.B.D + A.\overline{B}.\overline{D}) \\ &= \overline{C} + \overline{A}.B.D + A.\overline{B}.\overline{D} \end{aligned}$$

4. Simplify the following Boolean expressions as much as possible using basic Boolean identities:

- (a) $S(A, B, C) = \overline{A}.\overline{B}.C + B.C + A.C$
- (b) $T(A, B, C, D) = \overline{A}.B + A.B.D + A.\overline{B}.C.\overline{D} + B.C$
- (c) $U(A, B, C, D, \dots) = A + \overline{A}.B + \overline{A}.\overline{B}.C + \overline{A}.\overline{B}.\overline{C}.D + \dots$
- (d) $V(A, B, C) = A.B.C + \overline{A}.\overline{B}.C + \overline{A}.B.\overline{C} + A.\overline{B}.\overline{C}$

Solution

The Boolean algebraic expressions can be simplified as follows:

(a)

$$\begin{aligned}
 S(A, B, C) &= \overline{A}.\overline{B}.C + B.C + A.C \\
 &= (\overline{A}.\overline{B} + B)C + A.C \\
 &= (\overline{A} + B)(\overline{B} + B)C + A.C \\
 &= (\overline{A} + B)(1)C + A.C \\
 &= (\overline{A} + B)C + A.C \\
 &= (\overline{A} + B + A)C \\
 &= (1 + B)C \\
 &= (1)C \\
 &= C
 \end{aligned}$$

(b)

$$\begin{aligned}
 T(A, B, C, D) &= \overline{A}.B + A.B.D + A.\overline{B}.C.\overline{D} + B.C \\
 &= (\overline{A} + A.D)B + (A.\overline{B}.\overline{D} + B)C \\
 &= (\overline{A} + A)(\overline{A} + D)B + (A.\overline{D} + B)(\overline{B} + B)C \\
 &= (1)(\overline{A} + D)B + (A.\overline{D} + B)(1)C \\
 &= \overline{A}.B + B.D + A.C.\overline{D} + B.C \\
 &= \overline{A}.B + B.D + A.C.\overline{D} + (A + \overline{A})B.C \\
 &= (\overline{A}.B + \overline{A}.B.C) + B.D + A.C.\overline{D} + A.B.C \\
 &= \overline{A}.B(1 + C) + B.D + A.C.\overline{D} + A.B.C(D + \overline{D}) \\
 &= \overline{A}.B(1) + (B.D + A.B.C.D) + (A.C.\overline{D} + A.B.C.\overline{D}) \\
 &= \overline{A}.B + B.D(1 + A.C) + A.C.\overline{D}(1 + B) \\
 &= \overline{A}.B + B.D(1) + A.C.\overline{D}(1) \\
 &= \overline{A}.B + B.D + A.C.\overline{D}
 \end{aligned}$$

(c)

$$\begin{aligned}
 U(A, B, C, D, \dots) &= A + \overline{A}.B + \overline{A}.\overline{B}.C + \overline{A}.\overline{B}.\overline{C}.D + \dots \\
 &= A + \overline{A}(B + \overline{B}.C + \overline{B}.\overline{C}.D + \dots) \\
 &= (A + \overline{A})(A + B + \overline{B}.C + \overline{B}.\overline{C}.D + \dots) \\
 &= (1)(A + B + \overline{B}(C + \overline{C}.D + \dots)) \\
 &= A + (B + \overline{B})(B + C + \overline{C}.D + \dots) \\
 &= A + (1)(B + C + \overline{C}(D + \dots)) \\
 &= A + B + (C + \overline{C})(C + D + \dots) \\
 &= A + B + (1)(C + D + \dots) \\
 &= A + B + C + D + \dots
 \end{aligned}$$

(d) This expression cannot be simplified any further. Thus

$$V(A, B, C) = A.B.C + \overline{A}.\overline{B}.C + \overline{A}.B.\overline{C} + A.\overline{B}.\overline{C}$$